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CHAPTER ONE

INTRODUCTION OF STATISTICS

Definition of Statistics

Statistics is the science of data. It enables the collection, organization, presentation, analysis, and interpretation of numerical data.

Dr. A.L. Bowely defined “Statistics are numerical statement of facts in any department of enquiry placed in relation to each other”.

R.A. Fisher defined, “The science of statistics is essentially a branch of applied mathematics and may be regarded as mathematics applied to observational data”.

According to Croxton and Cowden, “Statistics may be defined as the science of collection, presentation, analysis and interpretation of numerical data”.

Origin and Development of Statistics

The word '**Statistics**' seems to have been derived from the Latin word 'status' or the Italian word 'statista' or the German word 'statistik' each of which means a 'political state'. In ancient times, the government used to collect the information regarding the population and 'property of wealth' of the country- the former enabling the government to have an idea of the manpower of the country and the latter providing it a basis for introducing new taxes and levies.

Major area of Statistics:

1. Descriptive Statistics
2. Inferential Statistics.

Descriptive Statistics

Descriptive statistics involves methods of organizing, picturing and summarizing information from data.

Inferential Statistics.

Inferential statistics involves methods of using information from a sample to draw conclusions about the population.

Uses and Importance of Statistics to Computer Students

Statistics is primarily used either to make predictions based on the data available or to make conclusions about a population of interest when only sample data is available. In both cases statistics tries to make sense of the uncertainty in the available data. When making predictions statisticians determine if the difference in the data points are due to chance or if there is asystematic relationship. The more the systematic relationship that is observed the better the prediction a statistician can make. The more random error that is observed the more uncertain the prediction. Statisticians can provide a measure of the uncertainty to the prediction. When making inference about a population, the statistician is trying to estimate how good a summary statistic of a sample really is at estimating a population statistic. For computer students, knowing the basic principles and methods in statistics could help them in doing their research work like comparing the speed of internet connection in different countries and the probability of how many times does each experience the same level of internet connection speed in a week, month or year. It could also be helpful in determining the best operating system to use. Whenever there is the need to compare data and know the best option that we should take statistics can give the answer.

Functions of statistics:

1. It simplifies much of figure.
2. It facilitates classification and comparison of data.
3. It helps in determining the relationships two or more phenomenon.
4. It helps in formulating and testing suitable hypothesis.
5. It helps a central management in formulating suitable future policy.
6. It helps in predicting future trends.

Limitations of Statistics:

1. Statistics deals with aggregate of items and not with individual item or measurement.
2. Statistics deals only with quantitative characteristics.
3. Statistical laws hold good only for the averages.
4. It plays only an auxiliary rule in summarizing a fact.
5. Statistics can be misused.

Characteristics of statistics:

The characteristics of statistics are given below:

1. Statistics should deal with aggregate of individual rather than with individual alone.
2. Statistics should be expressed as numerical figure.
3. Statistics should have the property of being varied by multiplicity of causes.
4. Statistics are collected or estimated should be reasonable of accuracy.

5. Statistics should be obtained by pre-determined purpose.
6. Statistics are collected in a systematic manner.

Importance of statistics:

The importance of statistics is given below:

1. Statistics of wealth and manpower are important for development and planning.
2. Statistics are invaluable in business and commerce.
3. Statistics helps the planner to estimate the revenue income and expenditure of the country.
4. Agriculture Statistics may play a key role in agriculture development.
5. In industry, statistics is widely used to provide quality control.
6. Statistics is usually used in education and psychology too.

Relation between Computer and Statistics:

Statistics is defined as the science of collecting, organizing, and interpreting numerical facts, which we call data. It is very important for a student of computer sciences. As Computer sciences also deals with organization and interpretation of numerical facts. In fact most of the principles of computer sciences are based on concepts of statistics.

The computer can process large amount of data quickly and accurately. For processing the large amount of data some of the important statistical packages that have been used are – SPSS, SAS, STATA, S-plus and MINITAB.

Population with example:

A population is the entire group of individuals, objects, or events with a common characteristic that is being studied. It is denoted by N.

Example: If we measure the heights and weights of a group of person then it is called population.

Populations are two types: (i) Finite population;

(ii) Infinite population.

Finite population: A Population is called finite population if its elements are countable. Example: Number of students in a university.

Infinite population: A population is called infinite population if its elements are not countable.

Example: Number of fishes in the Bay of Bengal.

Sample with example:

A representative part of population is called sample Or a part of the population is called **sample**.

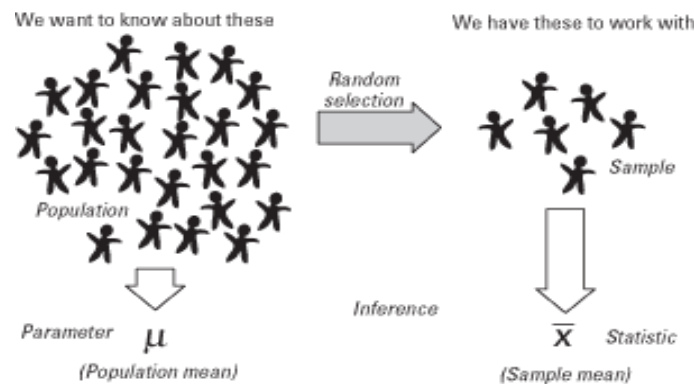
Example: If we measure the height and weight of IIUC student, then that of the CSE/EEE/ETE department student are sample.

Difference between a population and a sample

The main difference between population and sample are as follows:

Population	Sample
# The totality of all elements under the study or discussion is called population.	A part of the population is called sample.
# A population includes each element from the set of observations that can be made.	A sample consists only of observations drawn from the population.
# Population may be finite or infinite	Sample must be finite.
# Collecting data from every element of a population is not easy.	Collecting data is easy.
# All registered voters in our country	All registered voters in Chittagong district

** Figure of population and sample



Variable with example:

A variable is a measurable attribute of an object, event, or individual that can take on different values. Variables are represented by symbols (e.g., x, y, or z).

For example Age, weight, height, sex, income and expenses, country of birth, capital expenditure, class grades, eye colour and vehicle type are examples of variables.

Variables can be classified as **qualitative** (categorical) or **quantitative** (numeric).

Qualitative variable:

A variable which cannot be expressed as numerically is called **qualitative** (categorical) variable.

Examples: Hair color, gender, field of study, college attended, political affiliation, status of disease infection, sex, business type, eye colour, religion and brand.

Quantitative variable:

A variable which can be expressed as numerically then it is called quantitative variable.

Examples: Height, age, crop yield, GPA, salary, temperature, area, air pollution index (measured in parts per million), etc.

Quantitative variables can be further classified as **discrete** or **continuous**.

Discrete variable:

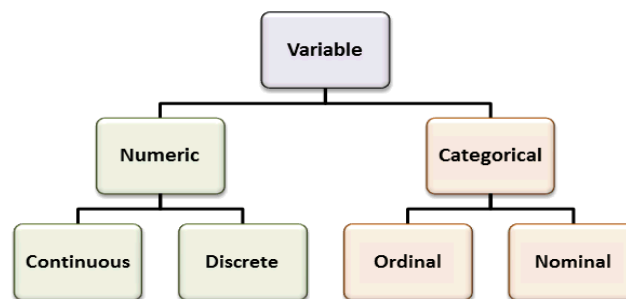
A variable is called discrete variable if it can take only isolated (Whole) values. Examples of discrete variables include the number of registered cars, number of business locations, and number of children in a family, all of which measured as whole units (i.e. ,1, 2, 3 cars).

Continuous variable:

A variable is called continuous variable if it can take any values between certain limits. i.e. Observations can take any value between a certain set of real numbers. Examples of continuous variables include height, time, age, and temperature.

N.B: Every continuous variable are discrete but every discrete variable are not continuous.

Figure of different types of variable:



Qualitative variable and quantitative variable:

Qualitative variable	Quantitative variable
Cannot be measured numerically	Can be measured numerically.
It is not countable	It is countable
Qualitative variable can be measure by using nominal and ordinal scale	Quantitative variable can be measure by using interval and ratio scale
The value of qualitative variable is generally discrete.	The value of quantitative variable is discrete and continuous
Algebraic expression meaningless	Algebraic expression meaningful

Skin colour, hair colour, religion, gender, merit, education, character are the example of qualitative variable.	GPA, age, weight, height, income, temperature etc are the example of quantitative variable.
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Discrete variable and continuous variable:

Discrete variable	Continuous variable
A variable is called discrete variable if it can take only isolated values	A variable is called continuous variable if it can take any values between certain limits
It is countable	It is measurable
The value of discrete variable may be finite or infinite	It must be infinite
Family member, student number are discrete variable	age, weight, height, salary etc are continuous variable

Constant:

Ans: A number that is not changing. It is usually denoted by a, b, c or d.

Example: Number of finger in a hand or leg.

Variable	Constant
A variable is always subject to change.	A constant will not change, ever.
It is denoted by X,Y,Z,U,V	It is denoted by a, b ,c or d
Variable are qualitative and quantitative	Constant has no such classification
age, weight, height, salary etc are variable	Total number of days in a week, Number of finger in a hand or leg are constant

Statistical Data

Ans: A set of observations obtained from a particular enquiry is called data.

Example: Income of workers or examination marks of a students.

According to sources data are:

1. Primary data;
2. Secondary data;

Primary data: Data collected by the investigator himself/ herself for a specific purpose.

Example: Data collected by different govt., public, private organizations, research bodies, research scholars, NGO's for their official records and research purpose from the field directly are primary data.

Secondary data: When an investigator uses the data which has already been collected by others, such data are called secondary data.

This data can be obtained from journals, report, Internet, Books/ Magazines, Newspapers, Office statistics the government statistics service, the office of national statistics, centre for applied social surveys.

Methods of collecting primary data:

1. Through interview
2. Through questionnaire
3. Through schedule
4. Through local agent.
5. Through observations
6. Through experimentation.

Difference between Primary and Secondary Data

Primary Data

1. Primary data are always original as it is collected by the investigator.
2. Suitability of the primary data will be positive because it has been systematically collected.
3. Primary data are expensive and time consuming.
4. Extra precautions are not required.
5. Primary data are in the shape of raw material.
6. Possibility of personal prejudice.

Secondary Data

1. Secondary data lacks originality. The investigator makes use of the data collected by other agencies.
2. Secondary data may or may not suit the objects of enquiry.
3. Secondary data are relatively cheaper.
4. It is used with great care and caution.
5. Secondary data are usually in the shape of readymade products.
6. Possibility of lesser degree of personal prejudice.

Measurement Scales:

1. Nominal Scale

- **Definition:** Categorizes data without any order or ranking.
- **Properties:** Only labels; cannot be ordered or ranked.
- **Mathematical Operations:** Only counting (frequency).

- **Examples:**
 - Gender: Male, Female, Other
 - Blood group: A, B, AB, O
 - Programming language: Python, C++, Java

2. Ordinal Scale

- **Definition:** Categorizes data with a meaningful order or ranking, but the intervals between ranks are not equal.
- **Properties:** Order matters, but differences are not measurable.
- **Mathematical Operations:** Comparisons (greater/less).
- **Examples:**
 - Customer satisfaction: Very Satisfied, Satisfied, Neutral, Dissatisfied, Very Dissatisfied
 - University ranking: 1st, 2nd, 3rd

3. Interval Scale

- **Definition:** Numeric data with equal intervals, but **no true zero** (zero does not mean absence).
- **Properties:** Differences are meaningful; ratios are not.
- **Mathematical Operations:** Addition, subtraction, mean, standard deviation.
- **Examples:**
 - Temperature in Celsius or Fahrenheit ($0^{\circ}\text{C} \neq$ no temperature)
 - Calendar years (2000, 2020, 2025 $\rightarrow$ 20 years apart)
 - IQ scores

4. Ratio Scale

- **Definition:** Numeric data with equal intervals **and a true zero point**.
- **Properties:** Differences and ratios are meaningful.
- **Mathematical Operations:** All arithmetic operations (add, subtract, multiply, divide).
- **Examples:**
 - Age, height, weight
 - Income, salary
 - Execution time of an algorithm (0 seconds = no time)
 - Number of errors in a program

Summary Table of Measurement Scales

Scale	Order	Equal Interval	True Zero	Examples
Nominal	✗	✗	✗	Gender, OS type
Ordinal	✓	✗	✗	Ranks, satisfaction level
Interval	✓	✓	✗	Temperature ($^{\circ}\text{C}$), calendar years
Ratio	✓	✓	✓	Age, weight, execution time

Questionnaire:

In any survey the information are collected according to some predetermined question. A set of question for any survey constitutes the questionnaire.

Or, A set of printed or written questions with a choice of answers, devised for the purposes of a survey or statistical study.

Characteristics of a good Questionnaire:

1. Questions worded simply and clearly, not ambiguous or vague
2. Write an introduction to the questionnaire
3. Question should be logically arranged.
4. Question should be as few as possible.
5. Question should be easy to understand.
6. Question should not be of multiple meaning.
7. Question should be capable of objective answer.
8. Question should be well designed.
9. Personal question should be avoided.
10. Write a descriptive title for the questionnaire

SAMPLE QUESTIONNAIRE:

Questionnaire on “Socio-Economic background and academic performance Of IIUC students”.

1.Students Name:	ID:
2. Semester	3. Department:
4. Mobile number (if possible):	5.Gender: (a) male, (b) Female
6. Age:.....	7. Religion: (a) Islam (b) Hindu (C)Others
8. Home District:	9. Number of brothers and sisters:.....
10. What is your serial number as a son/daughter of your parents? (a) First, (b) second, (c) third,(d) fourth (e) _____	11.Areas of your School at SSC or equivalent level : (a) city, (b) small town (c) village
12. Areas of your School at HSC or equivalent level : (a) city, (b) small town (c) village	13. Did you pass (a) SSC/HSC, (b) O-level/A-level,(c) Madrasa qualifications (d) others
14. SSC and HSC (or equivalent) GPA: SSC:_____ and HSC.....	15. Father's Education: (a) No education (b) Primary, (c) Secondary, (d) Higher Secondary (e) University.

16. Mother's Education: (a) No education (b) Primary, (c) Secondary, (d) Higher Secondary (e) university.	17. What is (or was) your father's occupation: (a) Agriculture (b) Business (c) Service (d) teacher and (e) Others.
18. What is (or was) your mother's occupation: (a) Housewife (b) Business (c) Service (d) teacher and (e) Others.	19. Your father and mother monthly income:.....
20. Economic status of your family:(a) Poor, (b) lower middle class (c) middle class, (d) higher middle class, (e)rich	21.Behind Choosing IIUC: (a) Islamic (b) Tuition (c) Safety and distance (d) Faculty (e)Good private university(f) Scholarship (g) others.
22. How you got information about IIUC: (a) Advertisement, (b) Faculty (c) friends/relatives, (d) internet	23.While attending university, I live: (a) In a hall (b) At home/ Family (c) Mess (d) Relative/friends (e) others
24. Approximately, what was your grade average in your final year of IIUC:.....out of 4.00.	25. Do you discuss your grades with your guardian? (a) Yes (b)No
26.Compared to my friends at university, I am, on an academic basis, performing:(a) Better (b) Same (c)Worse	27. How often did you miss classes: (a) Always (b) Sometimes (C) Never (d) Once every week?
28. Usually I study: (a) Library (b) Class (c) my room (d) If other please Specify:.....	29. Generally, I study.....hours daily.
30.Are you satisfied in IIUC academic system: (a) yes (b) No.	31. Do you seem that tuition fee is higher in IIUC? (a) yes (b) No.
32. What's your dream Career? (a) Subject related job(b) Teacher (c) Banker (d) service (e) others.	

THANK-YOU

Types of Questionnaire:

There are two types of questionnaire:

1. Unstructured question.
2. Structured question.

Unstructured question: Unstructured question are open-ended, that respondents can answer in their own words.

Example: What is your occupation?

Structured question: A question in which the respondents is given specific limited alternative responses and ask to choose one to his/her own view point.

Example: What is your occupation?

1. Agriculture
2. Business
3. Service
4. Teacher
5. Others.

Methods of organization of data:

1. Classification
2. Tabulation
3. Frequency distribution

Classification: Classification is the first statistical technique to condense the raw data. It is the process of arranging data in different groups or classes according to their affinities.

Tabulation: Tabulation is a logical and systematic arrangement of statistical data in rows and columns.

Frequency:

The number of times that a given value occurs into each group / class is known as frequency.

For example, if four students have a score of 80 in mathematics, and then the score of 80 is said to have a frequency of 4. The frequency of a data value is often represented by f .

Frequency distribution

Arrangement of observations according to frequencies is called frequency distribution.

Construction of a frequency distribution:

Following are the steps for the construction of a frequency distribution:

1. **Class limits/Range:** The class limits are the lowest and highest values that can be included in the class.
2. **Number of classes:** $K = 1 + 3.322 \log_{10} N$ where N is the number of observations,
3. **Class interval:** $C.I = \text{Range} / K$ where, $\text{Range} = H.V - L.V$.
4. **Class frequency/Tally:** The number of observations corresponding to the particular class is known as class frequency of that class.
5. **Mid value:** $(U.L + L.L) / 2$
6. **Cumulative frequency:** The number of observation for less than a certain upper limit of the class, if cumulation is done from top of the table.

Assignment-1: The following data refer to the ages of 60 employees of a firm.

33, 41, 21, 25, 36, 38, 35, 36, 35, 37, 42, 30, 35, 37, 36, 38, 30, 54, 40, 48, 15, 28, 51, 42, 25, 41, 30, 27, 42, 36, 28, 26, 37, 54, 44, 31, 36, 40, 36, 22, 30, 31, 19, 48, 16, 42, 32, 21, 22, 40, 43, 42, 39, 38, 37, 33, 49, 47, 46, 48.

- (i) Construct a frequency table with suitable class interval.
- (ii) Draw histogram, frequency polygon and ogive.

Tips: As like as class lecture.

Assignment-2: Monthly income (in Lac Tk.) of 30 firms in a certain area is given below:

20,19,10,13, 8,15,9,16,18,8,16,17,12,11,10,19,18,17,14,15,12,13,14,12,12.5,18.5,20,15,18,16.

- (i) Construct a suitable frequency distribution.
- (ii) Draw histogram and ogive.

Assignment-3: Tube light production (hourly) of 30 machines is given below:

9,20,19,10,12,15,9,16,18,8,16,17,12,11,10,19,18,17,14,15,12,13,14,12,12,18,20,15,18,15.

- (i) Construct a frequency distribution with suitable class interval.
- (ii) Draw ogive curve.

Example 6

The number of calls from motorists per day for roadside service was recorded for the month of December 2003. The results were as follows:

28	122	217	130	120	86	80	90	120	140
70	40	145	187	113	90	68	174	194	170
100	75	104	97	75	123	100	82	109	120
81									

Set up a frequency table for this set of data values.

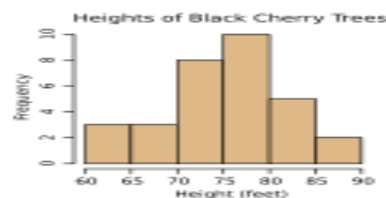
Graphical representation of statistical data

In addition to presentation of statistical data through tabular form, one can present the same through some visual aids refer to graphs and diagrams.

Types of Graphs and Diagrams:

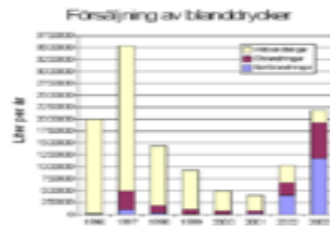
1. Histogram;
2. Bar diagram;
3. Frequency polygon
4. Pie diagram;
5. Scatter diagram;
6. Line diagram;
7. Ogive;
8. Steam and leaf plot.
9. Box plot

Histogram:



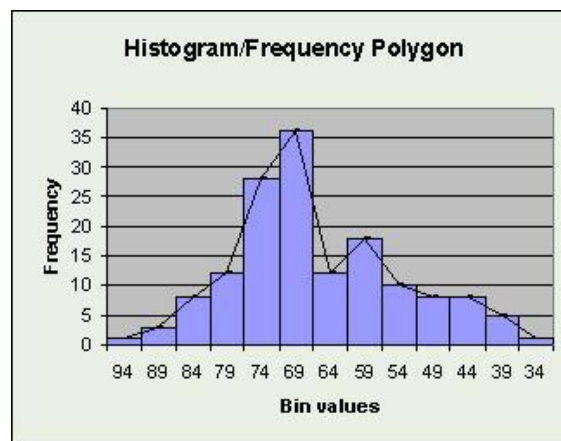
*A histogram consists of tabular frequencies, shown as adjacent rectangles, erected over discrete intervals (bins), with an area equal to the frequency of the observations in the interval.

Bar chart:

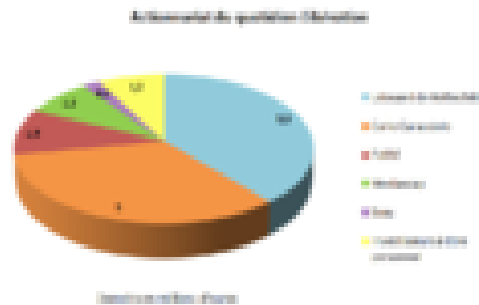


A bar chart is a chart with rectangular bars with lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally.

Frequency polygon

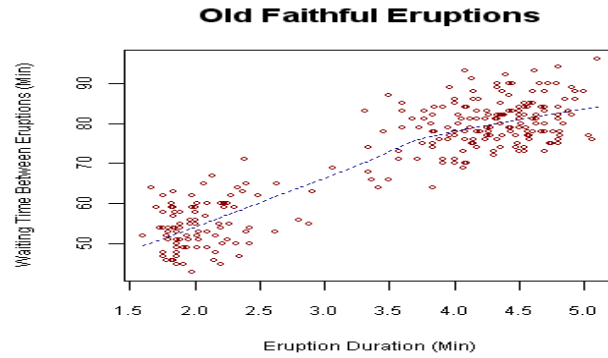


Pie chart:

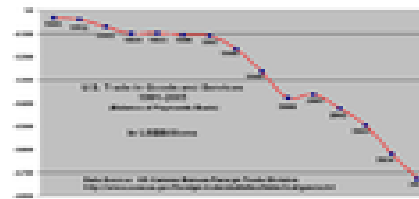


A pie chart shows percentage values as a slice of a pie.

Scatter diagram:

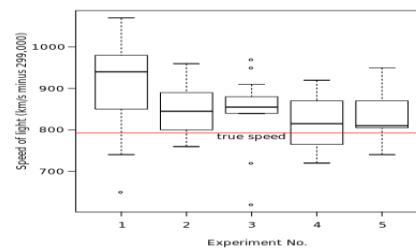


Line chart:



A line chart is a two-dimensional scatter plot of ordered observations where the observations are connected following their order.

Box plot



Cumulative frequency polygon or ogive

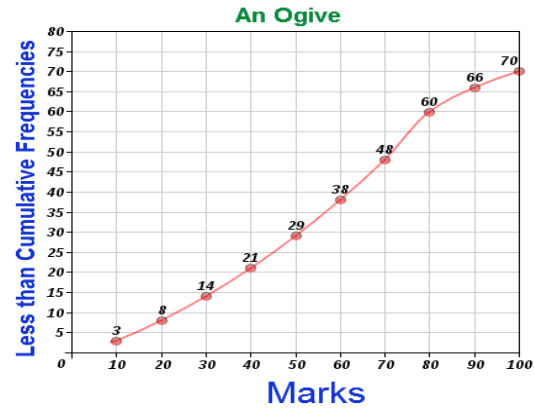


Figure: Ogive by using upper class interval

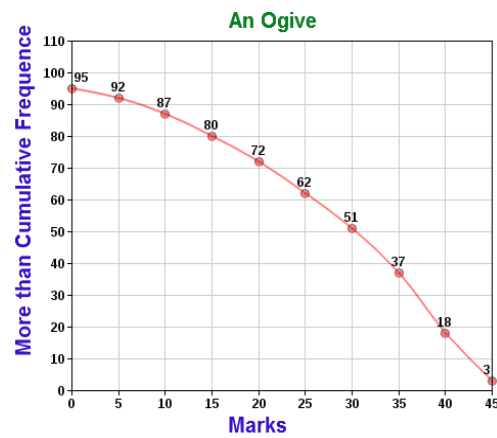


Figure: Ogive by using lower class interval

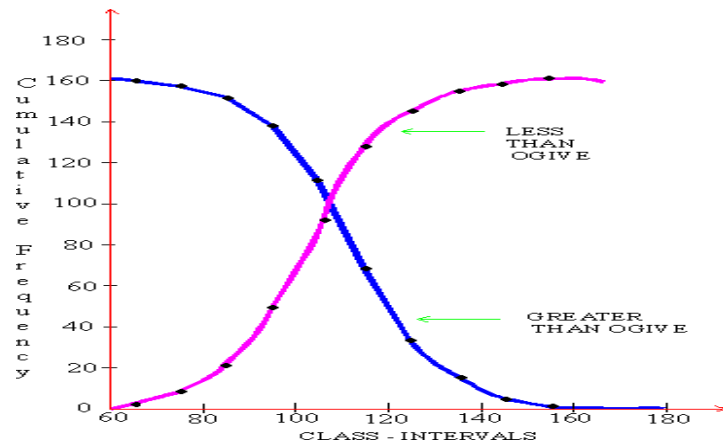


Figure: Ogive by using upper and lower class interval

The difference between graph and diagram:-

Diagram

- 1) Diagram can be drawn on plain paper and any sort of paper.
- 2) Diagram is more effective and impressive.
- 3) Diagram have everlasting effect.
- 4) Diagram cannot be used as median, mode etc.
- 5) Diagram can be represented as an approximate idea.

Graph

- 1) Graph can be drawn only on plain paper.
- 2) Graph is not more effective and impressive.
- 3) Graph doesn't have everlasting effect.
- 4) Graph can be used as median, mode etc.
- 5) Graph cannot be represented as an approximate idea.